

5th

Percentile and Quartiles

Percentage :- {1, 2, 3, 4, 5, 6}

$$\text{No of odd} = 3$$

$$\text{Percentage of odd} = \frac{3}{6} \times 100 = 50\%$$

Percentiles :- A percentile is a value below which certain percentage of observations lie.

{2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 9, 9, 10}

$$\text{Percentile of Value } x = \frac{\text{No. of values below } x}{n} \times 100$$

$$= \frac{11}{14} \times 100$$

$$= 78.57\% \text{ of value } x$$

Percentil Ranking

$$\Rightarrow \text{Value} = \frac{\text{Percentile}}{100} \times (n+1)$$

$$= \frac{25}{100} \times (14+1) = \frac{25}{100} \times 15 = \frac{15}{4}$$

$$= 3.75$$

② Quartiles

$$\approx 3.5$$

$$25\% = 1^{\text{st}} \text{ Quartile}$$

$$50\% = 2^{\text{nd}} \text{ Quartile}$$

$$75\% = 3^{\text{rd}} \text{ Quartile}$$

